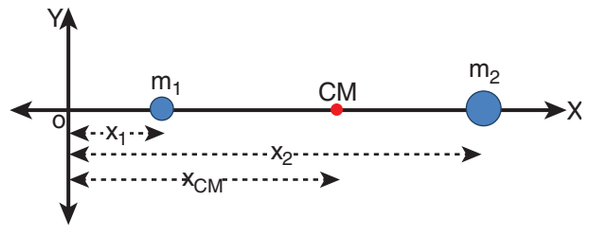
then, $x_{CM} = 0$ and the mass m_1 is found to be on the negative X-axis as shown in Figure 5.3(c). Hence, its position x_1 is negative, (i.e. $-x_1$).

$$0 = \frac{m_1(-x_1) + m_2 x_2}{m_1 + m_2}$$

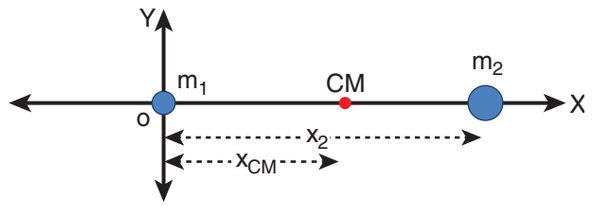
$$0 = m_1(-x_1) + m_2 x_2$$

$$m_1 x_1 = m_2 x_2$$

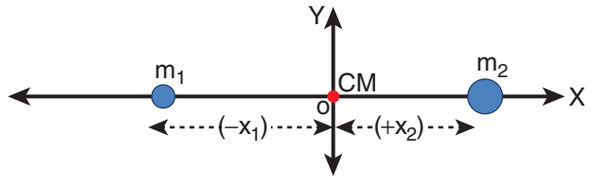
The equation given above is known as *principle of moments*. We will learn more about this in Section 5.3.3.



(a) When the masses are on positive X axis



(b) When the origin coincides with any one of the masses



(c) When the origin coincides with the center of mass itself

Figure 5.3 Centre of mass of two point masses determined by shifting the origin

EXAMPLE 5.1

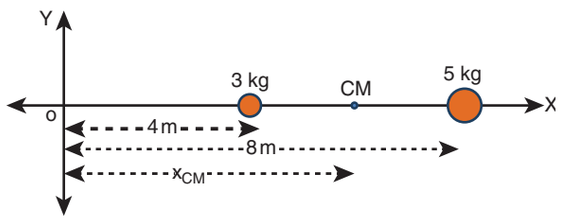
Two point masses 3 kg and 5 kg are at 4 m and 8 m from the origin on X-axis. Locate the position of centre of mass of the two point masses (i) from the origin and (ii) from 3 kg mass.

Solution

Let us take, $m_1 = 3$ kg and $m_2 = 5$ kg

(i) To find centre of mass from the origin:

The point masses are at positions, $x_1 = 4$ m, $x_2 = 8$ m from the origin along X axis.



The centre of mass x_{CM} can be obtained using equation 5.4.

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

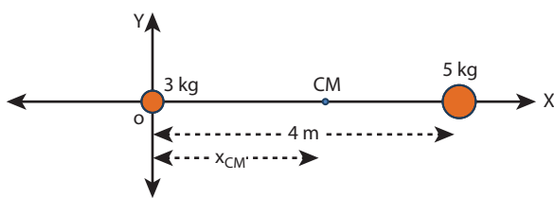
$$x_{CM} = \frac{(3 \times 4) + (5 \times 8)}{3 + 5}$$

$$x_{CM} = \frac{12 + 40}{8} = \frac{52}{8} = 6.5\,\text{m}$$

The centre of mass is located 6.5 m from the origin on X-axis.

(ii) To find the centre of mass from 3 kg mass:

The origin is shifted to 3 kg mass along X-axis. The position of 3 kg point mass is zero ($x_1 = 0$) and the position of 5 kg point mass is 4 m from the shifted origin ($x_2 = 4$ m).



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